

A Bulls App — The Living Universe

Build Brief for Expansion (handoff to build agent)

0. Non-negotiable constraints

- **Read-only, always.** No wallet connect, no custody, no commerce, no minting, no transaction signing anywhere in this product. Every surface only *observes and narrates* public on-chain data.
- **Bull Invaders is fully retired.** Remove all shooter-game code paths, assets, and references (power-ups, bosses, "the Herd," ship-select, EPOCH levels, Bullpen NFTs staking UI). Do not build on top of it, do not leave it reachable from any menu.
- **The existing particle-field engine is the foundation, not a prototype to discard.** The current build — the Three.js particle field, the

QUERY organism ("the Grey"), the voice layer, the read-only intelligence Worker, the full existing mode shell (`explore` , `intelligence` , `query` , `trickster` , `games` , `compare` , `what-if` , `sequences` , `evidence` , `replay`) — becomes **Galaxy Zero**, the first fully-built galaxy and the reference template every future galaxy must match in quality and feel. Do not rewrite it from scratch. Extend it.

- **Every already-stubbed mode gets built out fully in this pass, not simplified away.** `compare` , `what-if` , `sequences` , `evidence` , `replay` , and `trickster` already exist as defined interaction modes with their own hints in `MODE_HINT` — treat them as scoped features to implement, not as ideas to consolidate or drop.

1. Vision

A Bulls App is not a blockchain explorer. It's a living, explorable universe where on-chain activity is the terrain. Every ecosystem on Solana — every launchpad, every major protocol — is its own galaxy. Tokens are stars. Wallets are planets. Activity is weather. Death is visible. Birth is visible. You fly through it, ask it questions, and it answers out loud

in the voice of something that has been watching the whole time.

Long-term, this is not just a product — it's **infrastructure**. Other builders should eventually be able to build content, education tools, data products, and analysis tools on top of this universe the way people build on top of a map or an index, not just look at it.

2. What already exists and must be preserved / extended

Treat the current codebase as **Galaxy Zero**. Its components map onto the new architecture as follows — do not duplicate this logic elsewhere, generalize it:

Existing piece	New role
<code>particle-field.ts</code> <code>/ synthetic-universe.ts</code>	Becomes the shared rendering engine used by every galaxy, not just one

Existing piece	New role
<code>query-organism.ts</code> <code>/ alien-head-template.mjs</code> / <code>anatomy.ts</code>	The universal narrator ("the Grey") – same entity, reachable from any galaxy, always speaking only observed facts
<code>voice.ts</code>	Stays the shared narration layer
<code>field-os.ts</code>	Becomes the universe-level state machine: must gain a concept of "current galaxy" alongside its existing modes
<code>types.ts</code> <code>FieldMode</code> full set	Kept in full – see §5 for what each mode must actually do, per-galaxy
The read-only intelligence Worker	Becomes one of potentially several per-ecosystem data sources, unified behind one query interface
<code>games</code> mode	Reframed as a non-competitive education

Existing piece	New role
	layer — see §6

Do not re-architect the rendering or voice systems for this pass. The taxonomy work below is about *what data feeds the existing engine and how it's organized*, not about replacing the engine.

3. Cosmology — the taxonomy

This is the core design work. Every on-chain object maps to an astronomical object with a consistent, explainable rule — not an arbitrary skin.

Object	Maps to	Visual rule
Galaxy	An origin ecosystem/launchpad (pump.fun, Jupiter-launched tokens, ansem.io, Believe, etc.)	Fixed at launch-origin — a token's home galaxy never changes, even if it later trades elsewhere
Star	An individual token/mint	Brightness = liquidity/volume; color = health (green = healthy,

Object	Maps to	Visual rule
		red = bleeding, white/dying = collapsing)
Planet	A major holder wallet	Size = bag size relative to supply; orbital speed = trading frequency
Moon	An NFT collection tied to that token's ecosystem	Orbits the relevant planet or star
Asteroid belt	Liquidity pool / LP positions	Density = depth of liquidity
Comet	A large trade happening in near- real-time	The primary "alive, not archived" signal — must be visibly current, not batch-updated
Black hole	A rugged/dead token	Visibly pulls in and consumes nearby planets (holder exodus)

Object	Maps to	Visual rule
Supernova	A fast pump-and-death cycle	Leaves a permanent, discoverable scar/marker in that spot of the galaxy
Wormhole	A migration/bridge event (e.g. a pump.fun token graduating to Raydium, or a cross-chain bridge)	A literal traversable portal connecting two galaxies or two points in the same one
Ghost	A dormant/abandoned wallet, delisted token, or collapsed star	Renders as a translucent trace in the field, no longer active but still present. Querying a Ghost surfaces its underlying trade(s) plotted on an actual price/candlestick trading chart — not just the

Object	Maps to	Visual rule
		abstract particle visualization — so the claim "this thing traded here" is checkable against real price data, not just implied by the visuals

Galaxy boundary rule: a token's galaxy membership is permanent and determined by where it launched. Where it trades *today* is shown as wormhole trails leaving its home galaxy — this keeps every object's "address" stable while still being honest about where activity actually happens now.

4. Multi-galaxy architecture

- Galaxy Zero (the current field) should be generalized into "the template galaxy render pipeline," then instantiated once per ecosystem.
- Add a **galaxy selector** to the shell — reachable without leaving the living-universe feel (no

traditional dropdown/menu chrome breaking immersion; consider a starmap/overview view you can zoom out to).

- Each galaxy needs, at minimum: an origin/launch data source, a way to detect star birth (new token launch) and star death (rug/abandonment) in near-real time, holder/wallet data for planets, and OHLC/candlestick price data per token to back Replay, Ghost, and Evidence's chart views.
- Cross-galaxy objects (a wallet active in multiple ecosystems, a bridged token) should be visually and functionally consistent — same wallet = same planet identity wherever it appears.

Phase this in: ship the architecture (galaxy selector, generalized rendering pipeline, the taxonomy rules above) before populating every galaxy with real data. One additional fully-populated galaxy beyond Galaxy Zero is enough to prove the pattern generalizes.

5. Interaction modes

These already exist as stubs (`FieldMode` / `MODE_HINT`) and must be fully implemented,

working consistently across every galaxy, not just Galaxy Zero:

- **Replay** — play, pause, and step through chain time for a wallet, token, or region of a galaxy. This is the core timeline primitive everything else in this section builds on. Any specific trade surfaced during Replay must be viewable against an actual price/candlestick trading chart for that token, with the trade marked at its real point in price history — the particle field shows *where* it happened in the universe, the chart shows *what actually happened* to price.
- **Trickster (Create)** — sits directly on top of Replay, not a standalone content tool: the user scrubs Replay's timeline to select a bounded window of chain time, and Trickster composes that window into a narrated, evidence-backed story — the Grey's spoken commentary plus the visual route through the field — publishable as a replayable tour for others. Every claim in a Trickster story must cite the source event it came from; no narration without a receipt.
- **Compare** — place two wallets, tokens, or galaxies side by side with matched evidence, so claims about one are checkable against the other.

- **What-If** — simulate a hypothetical from observed evidence only (e.g. "what if this wallet had sold at the top"). Results must always be visually labeled as estimates, never presented as fact or blended visually with real observed data.
 - **Sequences** — surface bounded market sequences/patterns discovered across indexed events (e.g. pump → dump cycles, wash-trading loops).
 - **Evidence** — the sourcing/verification layer: for whatever is currently being viewed, shows original chain time, data coverage, and provenance — including a link to the same real trading chart used by Replay and Ghost, so any claim can be checked against actual price history, not just the visualization. This is the mode that makes every other mode's claims auditable.
 - **Ghost** — reveals dormant or dead objects (see §3) as translucent, queryable traces, so exploring a galaxy surfaces what died there, not just what's currently active.
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6. Education layer (formerly "Games")

Reframe `games` mode as **narrated on-chain education**: the Grey walks a user through a real example of a concept — what a rug looks like, how a launch graduates, what a wash-trading pattern looks like — using actual indexed events as the material. This is instructional, not competitive: no missions, scoring, or progression mechanics. It shares the same data and narration primitives as the other modes; it's a guided variant, not a separate system.

7. Infrastructure layer (future phase — describe now, do not build yet)

The long-term goal is that this universe becomes a platform others build on:

- A public API/SDK exposing the same indexed, read-only data that powers the galaxies.
- The rendering engine (galaxy/particle pipeline) exposed so third parties can build their own instruments, views, or entire sub-galaxies against the shared data spine.

- Third-party tours and educational content built through the same primitives used internally (§5, §6).

This section is context for architectural decisions now (e.g., keep the intelligence layer cleanly separated from the rendering layer so it can be exposed later) — it is **not** a deliverable for this build pass.

8. Explicit removals

- Delete all Bull Invaders game code, assets, and menu entry points (PIXI-based shooter, power-ups, bosses, "the Herd" hub, EPOCH terminology, Bullpen NFT staking/skin UI).
 - Do not reference or resurrect any of the above in new UI copy or code comments.
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9. Suggested delivery order

1. Generalize the existing rendering pipeline so it can be instantiated per-galaxy (no new galaxy data yet — prove Galaxy Zero still works identically through the generalized pipeline).

2. Implement the taxonomy mapping rules (§3) as a data layer, applied first to Galaxy Zero's existing data.
 3. Add the galaxy selector/starmap shell with Galaxy Zero as the only populated galaxy.
 4. Build out Replay fully, then Trickster on top of it (§5) – this is the highest-value pair since Trickster depends on Replay working first.
 5. Build out Compare, What-If, Sequences, Evidence, and Ghost (§5, §3) against Galaxy Zero's data.
 6. Stand up one additional galaxy end-to-end (recommend pump.fun for its clean birth → graduation/wormhole → death lifecycle), confirming every mode in §5 works identically there.
 7. Reframe `games` mode per §6.
 8. Remove all Bull Invaders code (can happen in parallel with step 1 – it's independent cleanup).
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10. Performance & scale constraints

- Assume tens of thousands of tokens per galaxy at maturity – use level-of-detail rendering (only

bright/active stars fully render at distance; detail resolves on approach) rather than attempting to render every object at once.

- Comets (real-time large trades) must feel live — avoid batch-only updates for this signal specifically, even if other data refreshes on a slower cycle.
- Replay must be able to scrub through historical chain time without re-fetching the entire window on every step — cache or pre-index the sequence being replayed.
- Maintain the existing mobile/reduced-motion/save-data handling already present in the field engine; extend it, don't bypass it, as new galaxies and modes are added.

11. Success criteria for this pass

- Galaxy Zero still works exactly as before, now running through the generalized pipeline.
- Replay and Trickster both work end-to-end: a user can scrub to a window of chain time and publish it as a cited, narrated tour.
- Compare, What-If, Sequences, Evidence, and Ghost are all functional, not placeholders.

- A second galaxy exists, populated with real data, using the same taxonomy and the same interaction modes.
- A user can move between galaxies without leaving the "living universe" feel (no jarring page-style navigation).
- No Bull Invaders code, assets, or references remain anywhere in the codebase or UI.
- The Grey still speaks only observed, sourced facts — no fabricated commentary, in any galaxy or mode.